

Exp No 3

To study Forward kinematics of 2 dof serial manipulators

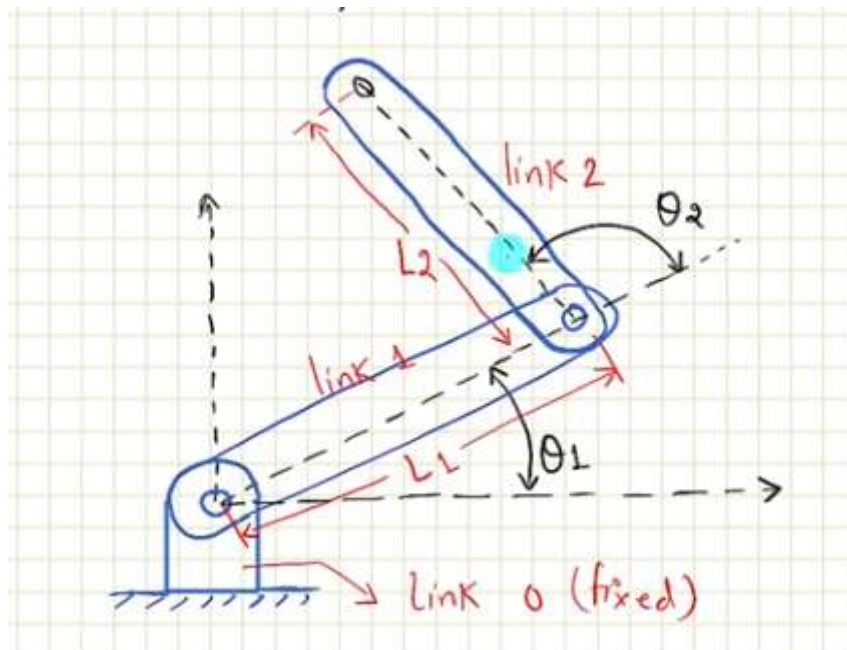
Aim: To study Forward kinematics of 2 dof serial manipulators

Instrument required:

Desktop PC with internet connection

Forward Kinematics:

Forward kinematics (FK) in robotics determines a robot's end-effector position and orientation (pose) in 3D space based on known joint parameters—such as angles for revolute joints or displacements for prismatic joints—using mathematical transformation matrices. It is essential for path planning, simulation, and controlling robot movements.



Step 1: Identify Home position:

D-H Parameter Table:

Frame	θ_i	d_i	a_i	α_i
0T_1	θ_1	0	L1	0
1T_2	θ_2	0	L2	0

$${}^0T_2 = Rot(Z, \theta_1).Trans(z, 0).Trans(x, L1).Rot(x, 0)$$

$${}^1_2T = Rot(Z, \theta_2). Trans(x, l_2)$$

$${}^0_2T = {}^0_1T \times {}^1_2T$$

Code for Matlab:

```
clc
clear

syms theta1 theta2 L1 L2
%theta1 = 0;
%theta2 = 0;

A = [cosd(theta1) -sind(theta1) 0 0;
sind(theta1) cosd(theta1) 0 0;
0 0 1 0;
0 0 0 1];

B = [1 0 0 L1;
0 1 0 0;
0 0 1 0;
0 0 0 1];

C = A*B

C =

[cos((pi*theta1)/180), -sin((pi*theta1)/180), 0, L1*cos((pi*theta1)/180)]
[sin((pi*theta1)/180), cos((pi*theta1)/180), 0, L1*sin((pi*theta1)/180)]
[ 0, 0, 1, 0]
[ 0, 0, 0, 1]

D = [cosd(theta2) -sind(theta2) 0 0;
sind(theta2) cosd(theta2) 0 0;
0 0 1 0;
0 0 0 1];

E = [1 0 0 L2;
0 1 0 0;
0 0 1 0;
0 0 0 1];

F=D*E

[cos((pi*theta1)/180)*cos((pi*theta2)/180) - sin((pi*theta1)/180)*sin((pi*theta2)/180), -
cos((pi*theta1)/180)*sin((pi*theta2)/180) - cos((pi*theta2)/180)*sin((pi*theta1)/180), 0,
L1*cos((pi*theta1)/180) + L2*cos((pi*theta1)/180)*cos((pi*theta2)/180) -
L2*sin((pi*theta1)/180)*sin((pi*theta2)/180)]
[cos((pi*theta1)/180)*sin((pi*theta2)/180) + cos((pi*theta2)/180)*sin((pi*theta1)/180),
cos((pi*theta1)/180)*cos((pi*theta2)/180) - sin((pi*theta1)/180)*sin((pi*theta2)/180), 0,
L1*sin((pi*theta1)/180) + L2*cos((pi*theta1)/180)*sin((pi*theta2)/180) +
L2*cos((pi*theta2)/180)*sin((pi*theta1)/180)]
[ 0, 0, 1, 0]
[0, 0, 0, 1]
```

Simplified:

$$G = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Answer:

[1, 0, 0, L1 + L2]

[0, 1, 0, 0]

[0, 0, 1, 0]

[0, 0, 0, 1]

Result:

The image shows a MATLAB script in the editor and the Command Window output. The script defines the Denavit-Hartenberg matrix G for a 2-link planar robot arm with link lengths L1 and L2, and joint angles theta1 and theta2. The matrix G is a 4x4 matrix with the following structure:

```

1  clc
2  clear
3
4  syms theta1 L1 theta2 L2
5  theta1 = 0;
6  theta2 = 0;
7
8  A = [cosd(theta1) -sind(theta1) 0 0;
9       sind(theta1)  cosd(theta1) 0 0;
10      0              0          1 0;
11      0              0          0 1];
12
13  B = [1 0 0 L1;
14      0 1 0 0;
15      0 0 1 0];
16
17  G = A*B;
18
19  display(G)
20
21  >>

```

The Command Window shows the resulting matrix G:

```

[1, 0, 0, L1 + L2]
[0, 1, 0, 0]
[0, 0, 1, 0]
[0, 0, 0, 1]

```